



17-4PH Stainless Steel

UNS S17400

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17-4PH stainless steel is chromium-nickel-copper stainless steel used for applications requiring high strength, high toughness, good corrosion resistance and high fatigue resistance. 17-4PH stainless steel is a Martensitic, precipitation-hardening stainless steel. The material is solution annealed (condition A) by heating to 1900F and rapidly cooling. It can then be single or double aged at appropriate time and temperature to increase the tensile properties as shown in the table below.

17-4PH stainless steel is used in the production of oilfield tools such as rotors, utilized in nuclear components, aerospace parts and many applications requiring high strength and corrosion resistance. Specialty Steel Supply can offer this grade in round bar and heavy plate. We can cut to length or water jet cut to your exact requirements utilizing our 5-axis system. We have in-house capability to straighten, turn, grind, and polish.



Standards

■ **UNS S17400**

Characteristics

- **High fatigue resistance.**
- **Good resistance to corrosion, similar to Grade 304.**
- **High strength.**
- **Readily weldable.**

Physical Properties

- **Density: 0.280 lb/in³ (7.8 g/cm³)**
- **Melting Point: 2560–2625°F (1404–1440°C)**
- **Modulus of Elasticity: 28.5x10⁶psi (197 GPa)**

Chemical Composition

As per AMS 5604

	Carbon C	Manganese Mn	Sulfur S	Phosphorus P	Silicon Si
MIN	-	-	-	-	-
MAX	0.07	1.00	0.03	0.04	1.00

Mechanical Properties

Mechanical properties as shown in ASTM A564

Condition	Tensile (min) KSI	Yield (min) KSI	Elongation (min) %	Reduction of Area (min) %	Hardness Brinell (Rockwell C)
Condition A	185	160	3%	-	-
H900	190	170	10%	35%	420 (C44)
H925	170	155	10%	35%	409 (C42)
H1025	155	145	12%	40%	352 (C38)
H1075	145	125	13%	45%	341 (C36)
H1100	140	115	14%	45%	332 (C35)
H1150	135	105	16%	50%	311 (C33)
H1150M	125	85	22%	68%	277 (C27)

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DATASHEET (PDF)

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Stainless and Alloys

We collaborate to help you exceed requirements by bringing your best ideas forward and eliminating non-conformance.

Specialty Steel Supply Inc. is one of the largest distributors of austenitic, precipitation hardened, duplex and martensitic stainless steels. We have a wide range of products and types of stainless steel round bar and our well trained staff offers many years of experience in all aspects of the steel industry. Our ability to procure and source metals from all parts of the world allows us to stay competitive in a global marketplace. We are committed to developing and servicing just in time inventory programs.

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Specialty Steel Supply

19201 Circle Lake Drive
Pinehurst, Texas 77362

(281) 821-7111

info@specialtysteelsupply.com

Business Hours

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Sunday - Closed

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